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Mixes and Medleys

January 2025 Problem 1 On this problem, no justification is required; only the answers will be graded.

- (a) Let R be a (not necessarily commutative) ring, and let M be a right R -module. Let A be the annihilator of M in R . Is A necessarily a two-sided ideal of R , necessarily a one-sided ideal of R , or not necessarily either of these things?
- (b) Let H be the subgroup of S_5 generated by the element $(12)(345)$. Then H acts on S_5 by left multiplication. How many orbits are there of this action?
- (c) Let $\mathbb{H}$ be the quaternion algebra $\mathbb{R} + \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$, and let A be its subalgebra $\mathbb{R} + \mathbb{R}i$ (so A is isomorphic to the complex numbers). What is the dimension of $\mathbb{H} \otimes_A \mathbb{H}$ as a real vector space?
- (d) True or false: if A is an $n \times n$ orthogonal matrix, then all n eigenvalues are either 1 or -1 .
- (e) Give an example of a prime ideal in the ring $\mathbb{C}[x, y, z]/(xy, xz, yz, x^2, z^2)$ which is not a maximal ideal.

August 2024 Problem 1 On this problem, no justification is required; only the answers will be graded.

- (a) The symmetric group on 5 letters S_5 acts on the set $\{1, \dots, 5\} \times \{1, \dots, 5\} \times \{1, \dots, 5\}$ by the diagonal action $\sigma(a, b, c) = (\sigma(a), \sigma(b), \sigma(c))$. How many orbits are there of this action?
- (b) The vector space $\mathbb{R}^3$ is a module for the ring of matrices $M_3(\mathbb{R})$. Let v be a nonzero vector in $\mathbb{R}^3$. What is the dimension of the annihilator of v in $M_3(\mathbb{R})$?
- (c) Let $\mathbb{F}_5$ be the field with 5 elements. How many ideals does the ring $\mathbb{F}_5[T]/(T^2 + 1)$ have?
- (d) Give an example of an abelian subgroup of $\mathrm{GL}_2(\mathbb{C})$ that is not contained in the group of diagonal matrices.
- (e) Give an example of a simple ring which is not a division ring.

January 2024 Problem 1 On this problem, no justification is required; only the answers will be graded.

- (a) Give an example of a simple ring which is not a division ring.
- (b) Let $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $Ae_1 = e_1 + e_2$, $Ae_2 = e_2$, $Ae_3 = e_3$. Let V be the vector space of linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$ which commute with A . What is $\dim_{\mathbb{R}}(V)$?

- (c) Give an example of a group G and a subgroup H , and a subgroup K of H , such that K is normal in H , and H is normal in G , but K is not normal in G .
- (d) Give an example of an inseparable field extension.
- (e) Give an example of a pair (R, M) , where R is a ring and M is an R -module that is not Noetherian.

August 2023 Problem 1 On this problem, no justification is required; only the answers will be graded.

- (a) Give an example of a field K , a finite algebraic field extension L/K , and a finite-dimensional K -algebra A such that A has no zero divisors but $A \otimes_K L$ does have zero divisors.
- (b) Give an example of a zero divisor in the $A \otimes_K L$ you provided above.
- (c) Give two elements x, y of the algebra of quaternions over $\mathbb{R}$ such that x and y both have norm 4 and x and y do not commute.
- (d) Give an example of a ring with exactly three ideals (including the zero ideal and the ring itself).
- (e) Give an example of a localization of $\mathbb{Z}$ which is not of the form $\mathbb{Z}[1/n]$ for any integer n .

January 2023 Problem 1 On this problem, only your answer will be graded, no justification is necessary.

- (a) Let R be a commutative ring and take $f, g \in R$. Consider the homomorphism $R \rightarrow R/(f) \otimes_R R_g$. Describe its kernel in terms of f and g .
- (b) Consider the homomorphism

$$\phi: (\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z}) \rightarrow \mathbb{Z}/5\mathbb{Z}: (a, b) \mapsto a + b.$$

(Note that $a + b$ makes sense as an element of $\mathbb{Z}/5\mathbb{Z}$.) The kernel $\ker \phi$ is a finite abelian group. What is its isomorphism class. (That is, its canonical form, such as $(\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z})$.)

- (c) True or False: if α and β are algebraic numbers such that $\mathbb{Q}(\alpha)$ is a Galois extension of $\mathbb{Q}$, and $\mathbb{Q}(\alpha, \beta)$ is a Galois extension of $\mathbb{Q}(\alpha)$, then $\mathbb{Q}(\alpha, \beta)$ is a Galois extension of $\mathbb{Q}$.
- (d) True or False: For any non-zero $f \in \mathbb{C}[x]$, any finitely-generated torsion-free module over $\mathbb{C}[x, \frac{1}{f}]$ is free.
- (e) True or False: For any n , any ideal $I \subset \mathbb{C}[x_1, \dots, x_n]$ can be generated by (at most) n elements.

January 2023 Problem 5 Let p be a prime number and let G be the group of 2×2 matrices of the form

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix}, \quad a \in \mathbb{F}_p, \quad b \in \mathbb{F}_p^\times.$$

Fix a dimension n , and let $\rho: G \rightarrow \mathrm{GL}_n(\mathbb{C})$ be a homomorphism. Put

$$x = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad y_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \quad (b \in \mathbb{F}_p^\times).$$

- (a) Show that $\rho(x)$ is diagonalizable, and that its eigenvalues are p th roots of unity.
- (b) Suppose $v \in \mathbb{C}^n$, $v \neq 0$ is an eigenvector of $\rho(x)$ of eigenvalue λ . Prove that for any $b \in \mathbb{F}_p^\times$, $\rho(y_b)v$ is also an eigenvector of $\rho(x)$, and find its eigenvalue.
- (c) Let v be as in part (b). Show that $W = \mathrm{span}\langle \rho(y_b)v : b \in \mathbb{F}_p^\times \rangle$ is G -invariant (that is, $\rho(g)W \subseteq W$ for all $g \in G$), and that, if $\lambda \neq 1$, $\dim W = p - 1$.

August 2022 Problem 1 On this problem, only your answer will be graded, no justification is necessary.

- (a) Let $K = \mathbb{Q}(2^{1/3})$. Is $K/\mathbb{Q}$ a Galois extension?
- (b) What is the radical of the ideal $(x^3 - x^2) \subset \mathbb{R}[x]$?
- (c) How many different (up to isomorphism) groups of order 49 are there?
- (d) Give an example of two non-zero abelian groups M, N such that $M \otimes_{\mathbb{Z}} N = 0$.
- (e) How many two-sided ideals are there in $M_2(\mathbb{C})$, the ring of 2×2 matrices with entries in $\mathbb{C}$?

January 2022 Problem 2 For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.
- (b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.
- (c) Every matrix in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ has finite order. (We write $\overline{\mathbb{F}}_p$ for the algebraic closure of the finite field $\mathbb{F}_p$.)
- (d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ is diagonalizable.

August 2021 Problem 1 On this problem, only the answer will be graded.

- (a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?
- (b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)
- (c) The Jordan form of an operator $A : \mathbb{C}^6 \rightarrow \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

January 2021 Problem 1 On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.
- (b) Give an example of two-sided ideal of the ring of 2×2 matrices over $\mathbb{Z}$ which is neither 0 nor the whole ring.
- (c) Give an example of an injective map $M \rightarrow N$ of $k[x]$ -modules (k a field) and another $k[x]$ -module R such that $M \otimes R \rightarrow N \otimes R$ is not injective.
- (d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\bar{q}$ of q to the localization R_p is prime, while the extension $\bar{q}'$ of q' to R_p is not prime.

January 2021 Problem 4, edited Let $F/\mathbb{Q}$ be a quadratic extension, and let f and g be two maps of $\mathbb{Q}$ -algebras from F to $M_2(\mathbb{Q})$. (**NB** in this problem we take algebra homomorphisms to be *unital*, meaning that $f(1) = g(1) = I$.)

- (a) Prove that there exists a matrix $X \in \text{GL}_2(\mathbb{Q})$ such that for all $\alpha \in F$, $Xf(\alpha)X^{-1} = g(\alpha)$.
- (b) Prove that there does not exist a choice of f whose image is contained in the set of upper triangular matrices.

August 2020 Problem 1 On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a non-diagonalizable matrix in $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (b) Give an example of a nontrivial proper normal subgroup of $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.
- (d) Give an example of a $\mathbb{Z}$ -module which is flat but not free.
- (e) Give an example of an exact sequence of abelian groups which is not split.

August 2019 Problem 1 On this problem, only the answers will be graded.

- (a) Let $R = M_2(\mathbb{Z})$ the ring of 2×2 matrices with entries in $\mathbb{Z}$. Give an example of a left ideal that is not a right ideal.
- (b) Give an example in $R = M_2(\mathbb{Z})$ of a proper, nonzero, two-sided ideal.
- (c) $\mathbb{Z}/100\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/35\mathbb{Z}$ is a finite abelian group. Compute its order.

August 2019 Problem 3 Let L/K be a Galois extension of fields with Galois group G . Let S be the subset of roots of unity in L ; that is, S is the set of elements $x \in L$ such that $x^k = 1$ for some positive integer k .

- (a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S .)
- (b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S .
- (c) Give an example of L/K such that the action of G on S is not faithful.
- (d) Suppose that the action of G on S is transitive. Prove that $\text{char}(K) = 2$.

January 2019 Problem 1 On this problem, only the answers will be graded.

- (a) Give an example of a simple ring which is not a commutative ring.
- (b) Give an example a ring R , R -modules M_1, M_2, M_3 , and N and a short exact sequence $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ such that the corresponding sequence $0 \leftarrow \text{Hom}(M_1, N) \leftarrow \text{Hom}(M_2, N) \leftarrow \text{Hom}(M_3, N) \leftarrow 0$ is not exact.
- (c) Let Φ be the map $\mathbb{Z}^3 \rightarrow \mathbb{Z}^2$ given by the matrix $\begin{bmatrix} 2 & 3 & 5 \\ 12 & -4 & 8 \end{bmatrix}$. Let M be the cokernel of Φ , which we recall is $\mathbb{Z}^2/\text{im}(\Phi)$. Compute the annihilator of M .

August 2018 Problem 5 Let A be a two-dimensional (unital) algebra over a field F . This means A is an associative, but not necessarily commutative, ring with a unit that contains F as a subring, such that the elements of F commute with the elements of A (that is, F is in the center) and A is two-dimensional as a vector space over F .

- (a) Show that A must in fact be commutative.
- (b) Show that if F is algebraically closed, then either $A \simeq F \times F$ or $A \simeq F[x]/(x^2)$.
- (c) Suppose $F = \mathbb{R}$. List (with a proof) all the possibilities for A , up to isomorphism.

August 2017 Problem 2 Let K be a field, and let A be an $n \times n$ matrix over K . Suppose that $f(x) \in K[x]$ is an irreducible polynomial such that $f(A) = 0$. Show that $\deg(f) \mid n$.

January 2017 Problem 5 Consider the ring $R = \mathbb{C}[x]$.

- (a) Describe all simple R -modules.
- (b) Give an example of an R -module that is indecomposable, but not simple. (Recall that a module is *indecomposable* if it cannot be written as a direct sum of non-trivial submodules.)
- (c) Consider R -modules $M = R/(x^3 + x^2)$, and $N = R/(x^3)$, and take their tensor product over R : $M \otimes_R N$. It is an R -module, and in particular a vector space over $\mathbb{C}$. What is its dimension over $\mathbb{C}$?
- (d) Let M be any R -module such that $\dim_{\mathbb{C}}(M) < \infty$, and let $N = R/(x^3)$ as before. Show that

$$\dim_{\mathbb{C}}(M \otimes_R N) = \dim_{\mathbb{C}} \operatorname{Hom}_R(N, M).$$

January 2016 Problem 5 Consider the group $\operatorname{GL}_n(\mathbb{Q})$ of invertible $n \times n$ matrices with rational coefficients. Suppose $G \subset \operatorname{GL}_n(\mathbb{Q})$ is a finite subgroup. Prove that every prime factor p of the order $|G|$ satisfies $p \leq n + 1$.

Linear Algebra

January 2025 Problem 4 Let V be a dimension 4 vector space over $\mathbb{C}$ and $\phi: V \rightarrow V$ an operator. Suppose that a dimension 2 subspace W of V is invariant in the sense that $\phi(W) \subset W$; then ϕ induces operators $W \rightarrow W$ and $V/W \rightarrow V/W$.

Suppose that the Jordan form of ϕ on both W and V/W is a single 2×2 Jordan block with eigenvalue 0. Find all possible Jordan forms of ϕ on V .

(For full credit, you need to provide examples that show your answers indeed can happen and prove that there are no other possibilities.)

August 2024 Problem 3 For an integer $n \geq 1$ consider the following $n \times n$ matrix:

$$A = \begin{pmatrix} 0 & & & 1 \\ & & 1 & \\ & & \ddots & \\ & 1 & & 0 \\ 1 & & & \end{pmatrix}.$$

Let V denote the set of all $n \times n$ matrices over $\mathbb{C}$ that are upper triangular and commute with A .

- (a) Show that V is a vector space over $\mathbb{C}$.
- (b) Find a basis for V .
- (c) Find the dimension of V .

August 2023 Problem 3 Let V be a four-dimensional real vector space with a positive definite inner product, and let $R: V \rightarrow V$ be an orientation-preserving automorphism that respects the inner product (that is, $\langle Rv, Rw \rangle = \langle v, w \rangle$ for all vectors $v, w \in V$).

- (a) Show that V decomposes as the direct sum of two subspaces V_1 and V_2 , each of dimension 2, each of which is preserved by R .
- (b) Show furthermore that V_1 and V_2 can be chosen to be orthogonal (that is, $\langle v_1, v_2 \rangle = 0$ for any $v_1 \in V_1, v_2 \in V_2$).

(NB in this context, “orientation-preserving” means $\det R > 0$.)

January 2022 Problem 4 Let V and W be vector spaces over a field k , not necessarily finite dimensional. As usual, we denote by $\text{Hom}_k(V, W)$ the space of linear maps from V to W , and by $V^\vee$ the dual of V , so that $V^\vee = \text{Hom}_k(V, k)$.

- (a) Construct a natural injective linear operator

$$\Phi: V^\vee \otimes_k W \rightarrow \text{Hom}(V, W),$$

and prove that it is indeed injective. (“Natural” means it should be independent of any arbitrary choices.)

- (b) Prove that the operator Φ is an isomorphism if V and W are finite-dimensional.
- (c) Prove a necessary and sufficient condition that a linear map $A \in \text{Hom}(V, W)$ is in the image of Φ . (The assumption that V and W are finite-dimensional does not apply to this part.)

August 2020 Problem 2 Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$; let A denote the transformation corresponding to M , and let B denote that corresponding to M^T .

- (a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\text{im}(B)$ with respect to the standard inner product (dot product).
- (b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\text{im}(B)$.

January 2020 Problem 4 Let V be a finite-dimensional vector space over a field. Let $A: V \rightarrow V$ be a linear transformation.

- (a) Show that there exists $B: V \rightarrow V$ such that $ABA = A$.
- (b) Show that there exists $C: V \rightarrow V$ such that $ACA = A$ and $CAC = C$.

(NB a matrix C like in part (b) is known as a “pseudo-inverse” to A .)

August 2019 Problem 4 This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an ***E*-matrix** if M has 1’s along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an *E*-matrix. Express the matrix A below as a product of *E*-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

January 2019 Problem 5 Let F be a field and n be a positive integer. Fix an $n \times n$ matrix S over F that is invertible and symmetric. Writing A^t for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F : A^t = SAS^{-1}\}.$$

Note that V is a vector space (you do not need to prove this). Find the dimension of V in terms of n .

August 2018 Problem 3 Denote by $M_n(\mathbb{R})$ the ring of all $n \times n$ matrices over $\mathbb{R}$.

- (a) Show that for any $A \in M_n(\mathbb{R})$, there exists a $B \in M_n(\mathbb{R})$ such that $AB = 0$ and $\text{rank}(A) + \text{rank}(B) = n$.
- (b) Prove or disprove that for any $A \in M_n(\mathbb{R})$, there exists $B \in M_n(\mathbb{R})$ such that $BA = AB = 0$ and $\text{rank}(A) + \text{rank}(B) = n$.

January 2017 Problem 2 Let $n > 0$ be an integer. Let F be a field of characteristic zero, V a vector space over F of dimension n , and $T: V \rightarrow V$ an invertible F -linear map such that $T^{-1} = T$. Denote by W the vector space of linear transformations from V to V that commute with T . Find a formula for $\dim(W)$ in terms of n and the trace of T .

August 2016 Problem 2 Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A: V \rightarrow V$, show that the following are equivalent.

- (i) There exists a linear transformation $P: V \rightarrow V$ such that $P^2 = I$ and $AP = -PA$.
 - (ii) There exists an invertible linear transformation $P: V \rightarrow V$ such that $AP = -PA$.
 - (iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$.
- (Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)

January 2015 Problem 4 Let F be a field, and let V be a finite-dimensional vector space over F that has dimension $n \geq 1$. Let q be a nonzero element of F such that $q^i \neq 1$ for $1 \leq i \leq n$. Let $X: V \rightarrow V$ and $Y: V \rightarrow V$ be two F -linear transformations such that $XY = qYX$. Show that XY is nilpotent.

August 2014 Problem 3

- (a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T: V \rightarrow V$ be a linear transformation. Show that T has an eigenvector.
- (b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T: V \rightarrow V$ such that T does not have an eigenvector.
- (c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

- (d) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U .

January 1994 Problem 4 Let X be a subspace of $M_n(\mathbb{C})$, the $\mathbb{C}$ -vector space of all $n \times n$ complex matrices. Assume that every nonzero matrix in X is invertible. Prove that $\dim_{\mathbb{C}}(X) \leq 1$.

January 1983 Problem 1 Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T: V \rightarrow V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x - \alpha)^n$, where $n = \dim V$ and α is some complex number. (**NB** As an extra challenge, try doing this without JNF.)

Group Theory

January 2025 Problem 2 Let $n \geq 2$ be an integer, and $p \geq 3$ be a prime. Denote by B the group of invertible $n \times n$ upper triangular matrices with entries in $\mathbb{Z}/p\mathbb{Z}$, and let U be the subgroup of B consisting of upper triangular matrices whose diagonal entries are all 1.

- (a) What is the order of the quotient group B/U as a function of p and n ?
- (b) Give an example of a nontrivial element in the center of U .
- (c) Prove that every nontrivial subgroup of U has nontrivial center.
- (d) Compute the center of B .

August 2024 Problem 4, August 2016 Problem 1 Let G be a group. By a maximal subgroup of G , we mean a subgroup $M \neq G$ such that the only subgroups containing M are M and G .

- (a) Describe all the maximal subgroups of the dihedral group of order $2p$ where p is an odd prime. How many are there?
- (b) Show that if a finite group G has only one maximal subgroup, then G is cyclic.
- (c) Show that if a maximal subgroup $M \triangleleft G$ is normal, then the index of M in G is finite and prime.

January 2024 Problem 3 Let G be a group acting transitively on a set X . Let $H \subset G$ be a normal subgroup. Then H naturally acts on X , but the action need not be transitive. Prove that all orbits of H on X have the same cardinality. (You may assume G and X are finite if you like, but this is not necessary!)

August 2023 Problem 5, edited By S_n we mean the symmetric group on n elements.

- (a) Let G be a subgroup of S_n . Suppose the action of G on $\{1, \dots, n\}$ is transitive. (**NB** such a subgroup is called a “transitive” subgroup.) Prove that for any $x, y \in \{1, \dots, n\}$, the stabilizer of x in G and the stabilizer of y in G are conjugate subgroups of G .
- (b) Prove that if G is a transitive abelian subgroup of S_n , then $|G| = n$.
- (c) Give an example of an integer n and a subgroup G of S_n such that G is transitive and abelian but not cyclic.
- (d) For which integers n does S_n contain a transitive abelian subgroup G that is not cyclic? (Justify your answer.)

January 2023 Problem 2 We say a group G has *big finite quotients* if for every $k > 0$, there is a normal subgroup N of G such that G/N is finite and has order at least k .

- (a) Give (with proof) an example of an infinite group that does *not* have big finite quotients.
- (b) Show that $\mathrm{SL}_2(\mathbb{Z})$ (the group of 2×2 integer matrices with determinant 1) has big finite quotients.

August 2022 Problem 2 Let $\mathbb{F}_p$ be the finite field with p elements; here p is a prime number. Let $V = \mathbb{F}_p^2$, and recall that $G = \mathrm{GL}_2(\mathbb{F}_p)$ is the group of invertible linear transformations on V . G acts on V in the usual way (by multiplication).

- (a) Describe the orbits of this action.
- (b) Describe the stabilizer in G of the vector

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in V.$$

- (c) Consider now the action of G on $V \times V$ (acting independently on each of the two vectors). How many orbits does this action have?
- (d) What is the cardinality of G ? Remember to justify your answer.

January 2022 Problem 1 On this problem, only the answer will be graded. Consider the symmetric group S_4 .

- (a) What is the order of S_4 ?
- (b) What is the order of its center $Z(S_4)$?
- (c) What is the order of its commutator subgroup $[S_4, S_4]$?
- (d) Is S_4 a simple group?
- (e) Is S_4 a solvable group?
- (f) How many conjugacy classes does S_4 have?

August 2021 Problem 3 Let G be a finite group. For given $n > 1$, we say that G *admits roots of degree n* if, for every $x \in G$, there exists $y \in G$ such that $y^n = x$.

- (a) Show that if n is coprime to $|G|$, G admits roots of degree n .
- (b) Conversely, prove that if G admits roots of degree n , then n is coprime to $|G|$.

January 2020 Problem 5

- (a) Find a nontrivial finite subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (b) Find an infinite, abelian, but non-cyclic subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (c) Find a proper, normal subgroup of $\mathrm{GL}_2(\mathbb{Z})$ of finite index.

August 2019 Problem 5 Let G be the group of 2×2 invertible upper triangular matrices over the field $\mathbb{F}_p$.

- (a) Prove that G has only one subgroup of order p , and that this subgroup is normal. (Hint: first show that an element of order p must have 1's on the diagonal.)
- (b) Prove that G is solvable by exhibiting a homomorphism f from G to an abelian group A such that the kernel of f is also abelian.
- (c) Prove that if $p \neq 2$, G is not nilpotent.

January 2019 Problem 2 If G is a group and $x, y \in G$, then we let $\langle x, y \rangle$ denote the subgroup of G generated by x and y . For $x \in G$, define $\mathrm{Cyc}_G(x) := \{y \in G : \langle x, y \rangle \text{ is cyclic}\}$.

- (a) Show by example that $\mathrm{Cyc}_G(x)$ need not be a subgroup of G .
- (b) Let $\mathrm{Cyc}(G) = \bigcap_{x \in G} \mathrm{Cyc}_G(x)$. Show that $\mathrm{Cyc}(G)$ is a subgroup of G .
- (c) Show that $\mathrm{Cyc}(G)$ lies in the center of G .
- (d) Assume now that G is finite. Show that $\mathrm{Cyc}(G)$ is cyclic.

August 2018 Problem 2 For a finite group G , denote by $s(G)$ the number of subgroups of G .

- (a) Show that $s(G)$ is finite.
- (b) Show that if H is a nontrivial normal subgroup of G , then $s(G/H) < s(G)$.
- (c) Show that $s(G) = 2$ if and only if G is cyclic of prime order.
- (d) Show that $s(G) = 3$ if and only if G is cyclic of order p^2 for a prime p .

January 2018 Problem 4, modified Let G be a finite group. Denote by $\mathrm{Aut}(G)$ the group of automorphisms of G , and by $Z(G) \subset G$ the center of G .

- (a) Let $\mathrm{Inn}(G) \subset \mathrm{Aut}(G)$ be the subgroup consisting of automorphisms coming from conjugation, i.e. automorphisms of the form $x \mapsto gxg^{-1}$ for some $g \in G$. Prove that $\mathrm{Inn}(G)$ is isomorphic to $G/Z(G)$.
- (b) Show that if $G/Z(G)$ is cyclic, then G is abelian.

- (c) Show that if $\text{Aut}(G)$ is cyclic, then G is abelian.
- (d) Show that if G is abelian, then $x \mapsto x^{-1}$ is an automorphism of G . (**NB** in fact this is an if and only if.)
- (e) Deduce that there is no group G such that $\text{Aut}(G)$ is a nontrivial cyclic group of odd order.

August 2017 Problem 3 What is the smallest n such that the 3-Sylow subgroup of S_n is non-abelian? (You may use the Sylow theorem that all Sylow subgroups are conjugate, so that one 3-Sylow subgroup is non-abelian if and only if they all are.)

January 2017 Problem 1 Give an example of each of the following.

- (a) A group G with a normal subgroup N such that G is not a semidirect product $N \rtimes G/N$.
- (b) A finite group G that is nilpotent but not abelian.
- (c) A group G whose commutator subgroup $[G, G]$ is equal to G .
- (d) A non-cyclic group G such that every Sylow subgroup of G is cyclic.
- (e) A transitive action of S_3 on a set X of cardinality greater than 3.

January 2016 Problem 4 Let D_k be the dihedral group of order $2k$ where $k \geq 3$.

- (a) Show that the number of automorphisms of D_k is equal to $k \cdot \varphi(k)$, where φ is Euler's φ -function.
- (b) Let $\text{Aut}(D_k)$ denote the group of automorphisms of D_k . What is the structure of $\text{Aut}(D_k)$? Describe the group as explicitly as you can.

Field and Galois Theory

January 2025 Problem 3 Let F be a field, $K \supset F$ be a finite extension of F , and let a be an element of K .

- (a) Suppose the degree $[K : F]$ is relatively prime to 6. Show that $F(a) = F(a^3)$.
- (b) Suppose the degree $[K : F]$ is not divisible by 3. Either prove that we still have $F(a) = F(a^3)$, or show by example that this is false.

August 2024 Problem 5 Denote by ζ_m a primitive m th root of unity, i.e. a complex number such that $\zeta_m^m = 1$ but $\zeta_m^k \neq 1$ for any $k < m$.

- (a) Prove that, for any m , the field extension $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is Galois.
- (b) What is the Galois group of $\mathbb{Q}(\zeta_6)/\mathbb{Q}$?
- (c) Give an example of an m such that the Galois group of $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is not cyclic.

January 2024 Problem 2 The irreducible factors of $x^{11} - 1$ over $\mathbb{Q}$ are $f_1 = x - 1$ and $f_2 = x^{10} + x^9 + x^8 + \cdots + 1$. (You do not need to prove this.) Let K be the splitting field of f_2 .

- (a) What is the Galois group of K over $\mathbb{Q}$?
- (b) Let $a = \cos(2\pi/11) = (\exp(2\pi i/11) + \exp(-2\pi i/11))/2$. Show that $K/\mathbb{Q}(a)$ is a Galois extension and find its Galois group.
- (c) Show that $\mathbb{Q}(a)/\mathbb{Q}$ is a Galois extension and find its Galois group.

August 2023 Problem 2 Let p be an odd prime number. Prove that any two fields of order p^2 are isomorphic. (Do not just quote the theorem that there's a unique isomorphism class of fields of any prime power order; your job is to prove a case of that theorem!)

January 2023 Problem 4 Consider the finite field $\mathbb{F}_p$ where p is a prime number.

- (a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p .)
- (b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

August 2022 Problem 4 Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

- (a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.
- (b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.
- (c) Show that K has a single degree p inseparable extension.

January 2022 Problem 5 Let F be a field of prime characteristic p . Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

- (a) Find $[E : F]$.
- (b) Prove that E/F is a Galois extension.
- (c) Find the Galois group $\text{Gal}(E/F)$.

Hint: What is $(x + 1)^p - (x + 1)$ in characteristic p ?

August 2021 Problem 2 Let $n > 1$ be a square-free integer.

- (a) Let $\alpha \in \mathbb{C}$ be a fourth root of n , i.e. $\alpha^4 = n$. Prove that $K = \mathbb{Q}(\alpha)$ is a degree 4 extension of $\mathbb{Q}$.
- (b) Find (with proof) the Galois group of the normal closure of K (in $\mathbb{C}$) over $\mathbb{Q}$.

January 2021 Problem 5 Let $\mathbb{F}_p$ denote the field with p elements.

- (a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?
- (b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a .
- (c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

August 2020 Problem 3 Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\text{Gal}(L/K)$. Then σ is a linear transformation of the K -vector space L .

- (a) What is the characteristic polynomial of σ ?
- (b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0 .

- (c) What is the characteristic polynomial of σ acting on L^0 ?

January 2020 Problem 2

- (a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.
- (b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F .
- (c) Compute the Galois group $\text{Gal}(F(b)/F)$.

January 2019 Problem 4

- (a) Let E be the splitting field over $\mathbb{Q}$ of $e(x) = x^3 + 2x^2 + 3x + 4$, which has discriminant -200 . Find $\text{Gal}(E/\mathbb{Q})$.
- (b) Let F be the splitting field over $\mathbb{Q}$ of $f(x) = x^3 + 4x^2 + 7x + 6$, which has discriminant -72 . Find $\text{Gal}(F/\mathbb{Q})$.
- (c) Find $\text{Gal}(EF/\mathbb{Q})$.

August 2018 Problem 4

- (a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.
- (b) Let $K = \mathbb{F}_q(t)$, and let $L = \mathbb{F}_q(t^{1/p})$ where q is a power of p . Prove that there are no automorphisms of L fixing K .

January 2018 Problem 5 Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\text{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)

August 2017 Problem 4 Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K : \mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m . Show that $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

January 2017 Problem 4 This is a question about “biquadratic extensions” in two parts.

- (a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\text{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E .
- (b) Give an example of a field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$.

August 2016 Problem 5, modified Consider the field $K = \mathbb{Q}(\zeta_7)$. According to your computation above, $\text{Gal}(K/\mathbb{Q})$ is abelian and has even order, so has a normal subgroup of index 2. By the Fundamental Theorem of Galois Theory, this subgroup corresponds to a field $\mathbb{Q}(x) \subset K$ such that $[\mathbb{Q}(x) : \mathbb{Q}] = 2$, where x is some element of K . Compute an element x in terms of ζ_7 which generates this quadratic subfield.

January 2016 Problem 1 Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f .

- (a) Show that if $\text{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.
- (b) Is the converse statement true? That is, suppose $L = K(\alpha)$; must $\text{Gal}(L/K)$ be abelian?

August 2015 Problem 5 Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x . Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

August 2011 Problem 3 Let $K \subseteq F \subseteq E$ be fields with $E = F[\alpha]$ and with $\alpha^n \in F$ for some positive integer n . Suppose K contains a primitive n th root of unity, and let L be a field with $K \subseteq L \subseteq E$ and $L \cap F = K$.

- (a) If L is Galois over K , show that $L = K[\beta]$ for some element β with $\beta^n \in K$.
- (b) Show by example that L need not be Galois over K . (Hint. Take $n = 2$.)

Rings and Modules

Commutative rings

January 2025 Problem 5 Let A be a commutative Noetherian ring, and M an A -module. We say that M has finite length if there exists a positive integer n such that every chain

$$M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k$$

satisfies $k \leq n$. The smallest such n is then called the length of M .

- (a) Prove that if M has finite length, then so does every localization $M_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of A .
- (b) Now assume that A is local with maximal ideal $\mathfrak{m}$. Assume furthermore that the field $\kappa = A/\mathfrak{m}$ embeds into A as a subring. Prove that the length of M equals the dimension of M over κ via restriction of scalars.
- (c) Prove that the following converse of (a) holds if A has finitely many maximal ideals: under this assumption, M has finite length if $M_{\mathfrak{m}}$ has finite length for every maximal ideal $\mathfrak{m}$ of A . (You can use without proof the fact that an A -module M is zero if and only if $M_{\mathfrak{m}}$ is zero for every maximal ideal $\mathfrak{m}$ of A .)
- (d) Prove that the assumption in part (c) on the finiteness of the number of maximal ideals is necessary by providing an example of a ring A and an A -module M which is not of finite length, even though $M_{\mathfrak{m}}$ is of finite length for every maximal ideal $\mathfrak{m}$ of A . (Hint: you can find such an example where A is the ring of integers.)

August 2024 Problem 2 Let k be a field, and let A denote the ring $k[x, y]/(xy)$. Consider the A -module M which is the quotient of the free rank 2 A -module with basis e_1 and e_2 by the submodule generated by the element $ye_1 + xe_2$.

- (a) Prove that M is not a projective A -module. (Hint from the original test: find a submodule of M isomorphic to $k[x, y]/(x, y)$.) (**NB** There are other, more systematic, ways of doing this if you are comfortable with Tor.)
- (b) Let $f \in A$ denote the polynomial $x + y$, and let A_f denote the localization of A at the multiplicative subset $\{1, f, f^2, \dots\}$. Prove that $M \otimes_A A_f$ is free as an A_f -module.

January 2024 Problem 4, notation altered Let R be a commutative ring, and let A and B be commutative R -algebras.

- (a) Prove that if B is flat over R , and f is a non zero divisor in A , then $f \otimes 1$ is a non zero divisor in $A \otimes_R B$.
- (b) Prove that the flatness condition above is necessary by giving an example where part (a) fails if B is not flat over R .

August 2023 Problem 4 Two questions about homomorphisms of modules.

- (a) Let R be an integral domain and M an R -module. Suppose M is free of finite rank r . Exhibit an isomorphism of R -modules ϕ from M to $\text{Hom}_R(\text{Hom}_R(M, R), R)$ (and prove that ϕ is an isomorphism).
- (b) Give an example of an integral domain R and an R -module M such that M is not isomorphic as an R -module to $\text{Hom}_R(\text{Hom}_R(M, R), R)$.

January 2023 Problem 3, notation altered Let $S = \mathbb{C}[x, y, z]$. In this problem, you may use the fact that the maximal ideals $\mathfrak{m} \subset S$ are of the form $\mathfrak{m} = (x - a, y - b, z - c)$ for $(a, b, c) \in \mathbb{C}^3$.

Consider the sequence

$$S \xleftarrow{g} S^3 \xleftarrow{f} S^2,$$

where the maps f and g are given by the matrices

$$g = \begin{pmatrix} x & y & z \end{pmatrix}, \quad f = \begin{pmatrix} y & z \\ -x & 0 \\ 0 & -x \end{pmatrix}.$$

For a maximal ideal $\mathfrak{m}$, we tensor the sequence by $S/\mathfrak{m}$ and denote by $f_{\mathfrak{m}}$ and $g_{\mathfrak{m}}$ the induced maps

$$S \otimes_S (S/\mathfrak{m}) \xleftarrow{g_{\mathfrak{m}}} (S^3) \otimes_S (S/\mathfrak{m}) \xleftarrow{f_{\mathfrak{m}}} (S^2) \otimes_S (S/\mathfrak{m}).$$

- (a) For which maximal ideals $\mathfrak{m}$ is $f_{\mathfrak{m}}$ injective?
- (b) For which maximal ideals $\mathfrak{m}$ is $g_{\mathfrak{m}}$ surjective?
- (c) For which maximal ideals $\mathfrak{m}$ is $\ker g_{\mathfrak{m}} = \text{im } f_{\mathfrak{m}}$?

August 2022 Problem 3 Let R be a commutative ring. Suppose M and N are R -modules whose direct sum is a free module of finite rank, that is,

$$M \oplus N \simeq R^n$$

for some $n \geq 0$.

- (a) Show that M is finitely generated.
- (b) Show that M is finitely presented. (Recall that a module is *finitely presented* if it is isomorphic to the quotient of a finitely generated free module by a finitely generated submodule.)
- (c) Besides the two properties listed above, M also has another well-known property. What is it?

January 2022 Problem 3 Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

August 2021 Problem 4 Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A . Let $f: M \rightarrow N$ be a homomorphism of A -modules, and let $g: B \rightarrow A$ be a homomorphism of commutative rings.

- (a) “If f is injective, so is the induced morphism $M \otimes_A M \rightarrow N \otimes_A N$.”
- (b) “If f is surjective, so is the induced morphism $M \otimes_A M \rightarrow N \otimes_A N$.”
- (c) “If g is injective, so is the induced morphism $M \otimes_B M \rightarrow M \otimes_A M$.”

January 2021 Problem 3 Two questions about projective modules.

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

August 2020 Problem 4 Let R be a commutative ring.

- (a) Give an example of a free R -module M and an injective homomorphism $M \rightarrow M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)
- (b) If M_1 and M_2 are free of finite ranks r_1 and r_2 , show that a surjective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \geq r_2$.
- (c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \rightarrow M_2$ exists if and only if $r_1 \leq r_2$.

January 2020 Problem 3 Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
- (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .
- (a) Prove (i) or (ii).
- (b) Give an example of a ring with finitely many elements that has exactly three prime ideals.
- (c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

August 2019 Problem 2 This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I , denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

- (a) Prove that if I is primary, then $\sqrt{I}$ is prime.
- (b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x, y]$ primary? Be sure to carefully justify your answer.

January 2019 Problem 3, edited Let R be a commutative ring and $S \subseteq R$ a multiplicatively closed set.

- (a) Let M be an R -module. Prove that $S^{-1}M \cong S^{-1}R \otimes_R M$.
- (b) Give an example of a ring R , a multiplicatively closed subset $S \subseteq R$, and an R -module M where $S^{-1}M$ is flat (as an $S^{-1}R$ -module) and nonzero, but where M is not flat. (And prove that your example has those properties!)

August 2018 Problem 1, edited For this problem, your answer will be graded on correctness alone, and no justification is necessary.

- (a) Give an example of a ring R and a short exact sequence of R -modules $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ that is not split exact.
- (b) Give an example of a flat $\mathbb{Z}$ -module that is not free.
- (c) Describe all zero-divisors and all units in $\mathbb{Q}[x]/(x^2 - 1)$.

January 2018 Problem 1 For this problem, your answer will be graded on correctness alone, and no justification is necessary.

- (a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.
- (b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \rightarrow R_f$ is neither injective nor surjective.
- (c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

January 2018 Problem 2 Recall that a (left) zero-divisor in a ring R is an element a such that $ab = 0$ for some nonzero $b \in R$. Consider the rings

$$R_1 = \mathbb{C}[x]/(x^3) \quad \text{and} \quad R_2 = M_n(\mathbb{C}) \quad (n \times n \text{ matrices over } \mathbb{C}, \text{ where } n > 1).$$

- (a) Give an example of a nonzero zero-divisor in R_1 .
- (b) Give an example of a nonzero left zero-divisor in the ring R_2 .
- (c) Show that the set of zero-divisors in R_1 form an ideal, but the set of left zero-divisors of R_2 is not a left ideal.
- (d) Let S be a commutative ring. Prove that if the set of zero-divisors is an ideal I , then $I \subset S$ is a prime ideal.

August 2017 Problem 1, edited For this problem, your answer will be graded on correctness alone, and no justification is necessary.

Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

- (a) $M \otimes_{\mathbb{C}[x]} N$
- (b) $M \otimes_{\mathbb{C}} N$
- (c) $M \otimes_{\mathbb{C}[x^2]} N$.

August 2017 Problem 5, modified Let $B = \mathbb{Q}[x, y]/(xy)$.

- (a) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to x . This map makes B into a $\mathbb{Q}[t]$ -module. Show that B is *not* a finite $\mathbb{Q}[t]$ -module under this map.
- (b) Consider the map $\mathbb{Q}[t] \rightarrow B$ sending t to $x + y$. Show that this map makes B into a finite $\mathbb{Q}[t]$ -module.

- (c) Consider B as a $\mathbb{Q}[t]$ -module under the map from part (a). Is B a flat $\mathbb{Q}[t]$ -module? What about when considered as a $\mathbb{Q}[t]$ -module under the map from part (b).

January 2017 Problem 3 Let R be a commutative ring with unity. Show that a polynomial

$$f(t) = c_n t^n + c_{n-1} t^{n-1} + \cdots + c_0 \in R[t]$$

is nilpotent if and only if all of its coefficients $c_0, \dots, c_n \in R$ are nilpotent.

August 2016 Problem 4 Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

- (a) Show that R is generated as a $\mathbb{C}$ -algebra by $x + y$ and xy .
- (b) Show that the map $\mathbb{C}[u, v] \rightarrow R$ sending u to $x + y$ and v to xy is an isomorphism.
- (c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$ -algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

August 2014 Problem 4 Let R be a commutative ring and M an R -module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

- (a) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of R -modules, and let P be an associated prime of M . Prove that P is an associated prime of either M' or M'' .
- (b) Is the converse true? That is, if P is an associated prime of either M' or M'' , must it be the case that P is an associated prime of M ?

August 2013 Problem ? Let I and J be ideals in a commutative ring R .

- (a) Show that if $I + J = R$, then $I \cap J = IJ$.
- (b) Suppose that I and J are ideals in $\mathbb{C}[x]$, and suppose that $I + J = (x)$. Show that $(I \cap J)/IJ$ is 1-dimensional as a complex vector space, and moreover that it is isomorphic to $\mathbb{C}[x]/(x)$ as a $\mathbb{C}[x]$ -module.
- (c) On the other hand, for general commutative rings R , once $R/(I+J)$ is not trivial, the difference between $I \cap J$ and IJ can be large even if $R/(I \cap J)$ is small. Demonstrate this by showing that, if $R = \mathbb{C}[x, y]$, then there exists ideals I and J in R such that $I + J = (x, y)$ and the dimension of $(I \cap J)/IJ$ as a $\mathbb{C}$ -vector space is at least 100.

Noncommutative rings + applications to linear algebra

January 2024 Problem 5 Let $\mathbb{C}$ denote the field of complex numbers. Pick a nonzero $q \in \mathbb{C}$ such that $q^n \neq 1$ for all positive integers n . Let A denote the $\mathbb{C}$ -algebra with generators x, y subject to the relation $yx = qxy$.

- (a) Let V denote a nonzero A -module on which x and y are injective. Show that V has infinite dimension. (**NB** The module V is a $\mathbb{C}$ -vector space, and the problem is asking you to show that V has infinite dimension as a $\mathbb{C}$ -vector space.)
- (b) Give an example of a nonzero A -module on which x and y are injective.
- (c) Show that the following is a basis for the $\mathbb{C}$ -vector space A :

$$x^i y^j \quad i, j \geq 0.$$

August 2022 Problem 5, edited Let V be a vector space over $\mathbb{C}$ of dimension $n \geq 2$. Let

$$A: V \rightarrow V$$

denote a $\mathbb{C}$ -linear map with n mutually distinct eigenvalues. Prove that V contains one-dimensional subspaces $V_1, \dots, V_n$ such that

- (i) we have

$$V = \sum_{i=1}^n V_i$$

and if $i \neq j$, $V_i \cap V_j = \{0\}$ (**NB** these two conditions are equivalent in this case);

- (ii) $AV_i \subseteq V_i + V_{i+1}$ for $1 \leq i \leq n-1$ and $AV_n \subseteq V_n + V_1$; and
- (iii) V_i is not an eigenspace of A for $1 \leq i \leq n$.

August 2021 Problem 5 Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

- (a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$ -algebra homomorphism $\phi: A \rightarrow B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.
- (b) Prove that the homomorphism ϕ from part (a) is injective.
- (c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

January 2021 Problem 2 Let k be a field, and let $R = M_n(k)$, the non-commutative ring of $n \times n$ matrices over k .

- (a) Give examples of a simple left R -module M and a simple right R -module N .
- (b) Can you find another simple left R -module M' that is not isomorphic to M ? (Either find one or explain why there is none such.)
- (c) Compute $\dim_k(N \otimes_R M)$.

August 2020 Problem 5, edited Let $U: \mathbb{Z} \rightarrow \mathbb{Z}$ be the function defined by $U(n) = -n$, and let $T: \mathbb{Z} \rightarrow \mathbb{Z}$ be the function defined by $T(n) = n+1$. Denote by M the space of all functions from $\mathbb{Z}$ to $\mathbb{C}$. Then T and U can be thought of as elements of $\text{End}_{\mathbb{C}}(M)$ (the algebra of $\mathbb{C}$ -linear maps from M to M) by the operation of composition of functions. (For instance, if m is the function $m(n) = 3n$, then mU is the function sending n to $3(-n) = -3n$, mT is the function sending n to $3(n+1) = 3n+3$, and, taking V to be $2TU + 1$, mV is the function sending n to $2 \cdot 3(-n+1) + 3n = -3n + 6$.)

(NB there are a couple confusing things here. One is that the function composition mU and mT occurs *inside* the parentheses, i.e. $mU(n) = m(U(n))$, but the scalar multiplication occurs *outside* the parentheses, $m(2U)(n) = 2m(U(n))$. This is because the scalar multiplication is scalar multiplication in the complex vector space M , and the scalar multiplication there is scaling the function, not the input. Indeed, I can scale by any complex number, but the input cannot be scaled by an arbitrary complex number and remain an integer. The second thing is that the element written above as 1 in $2TU + 1$ is really the identity map on M , so $m(2TU + 1)(n) = 2mTU(n) + m(n)$.)

Let R be the subalgebra of $\text{End}_{\mathbb{C}}(M)$ generated by U and T ; that is, it is a complex vector space spanned by all linear maps obtainable by repeatedly composing copies of T and copies of U .

- (a) Show that R is not commutative.
- (b) M is an infinite-dimensional complex vector space which carries the structure of a right R -module by function composition. The space P_0 of constant functions is a 1-dimensional submodule of M . Give an example of a 1-dimensional submodule of M containing non-constant functions, and show that it is the only one. (NB This is *not* asking for a submodule that is generated by just one $f \in M$, it is asking for a submodule that is *also* a 1-dimensional complex vector space.)
- (c) There is a ring homomorphism $\phi: R \rightarrow \mathbb{C}$, called *augmentation*, defined by $\phi(T) = \phi(U) = 1$. Let I be the kernel of ϕ , which is a two-sided ideal. Prove that $M \otimes_R R/I = 0$.

January 2020 Problem 1 On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

- (a) Write down a nonzero element of R which is a zero-divisor.
- (b) Write down an element in the center of R which is not in $\mathbb{Z}$.

- (c) Write down an R -module which is not a free module.

January 2018 Problem 3 Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG , and let $T: FG \rightarrow FG$ be the F -linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T .